STAT lecture 8
Two-sample comparative hypothesis tests

Disclaimer:
Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with MATLAB examples

SDU

Testing hypotheses for two random samples (comparative tests)
Non-paired tests (two independent samples)

Model

$$X_{11}, \dots, X_{1n_1} \quad \text{iid} \quad X_{1i} \sim N(\mu_1, \sigma_1^2)$$
$$X_{21}, \dots, X_{2n_2} \quad \text{iid} \quad X_{2i} \sim N(\mu_2, \sigma_2^2)$$

Null hypothesis $H_0: \ \mu_1 - \mu_2 = \Delta_0$ against $H_1: \ \mu_1 - \mu_2 \neq \Delta_0$

Case 1: Hypothesis test for difference in means with known variances

Test statistic

$$Z_0 \overset{\text{def}}{=} \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

Test

reject H_0 for $|z_0| > z_{\alpha/2}$

p-value: $p = 2\left(1 - P(Z \leq |z_0|)\right) = 2(1 - \Phi(|z_0|))$

CI for $\mu_1 - \mu_2$:

$$\left[(\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right]$$

Case 2: Hypothesis test for difference in means with unknown but equal variances

Test statistic $$T_0 \overset{\text{def}}{=} \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{S_P\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

where the pooled (combined) estimate of the common, unknown variance is

$$S_P^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2} \sim \frac{\sigma^2}{n_1+n_2-2}\chi^2(n_1 + n_2 - 2)$$

and S_1^2, S_2^2 are the usual variance estimates from populations 1 and 2

Test reject H_0 for $|t_0| > t(n_1 + n_2 - 2)_{\alpha/2}$

p-value: $p = 2\left(1 - P(t(n_1 + n_2 - 2) \leq |t_0|)\right)$

CI for $\mu_1 - \mu_2$: $$\left[(\bar{X}_1 - \bar{X}_2) \pm t(n_1 + n_2 - 2)_{\alpha/2} S_P\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right]$$

Case 3: Hypothesis test for difference in means with unknown and unequal variances

Large sample approximate test statistic $$Z_0^* \overset{\text{def}}{=} \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} \sim N(0,1)$$

Paired tests (pairwise data from population 1 and 2)

Model (pairwise data) $(X_{1i}, X_{2i}), \quad i = 1, \ldots, n$

$$X_{1i} \sim N(\mu_1, \sigma_1^2)$$
$$X_{2i} \sim N(\mu_2, \sigma_2^2)$$

Paired tests have more statistical power than non-paired tests
Possible when each test object can be split in two parts for "treatments" 1 and 2

Difference model $D_i \stackrel{\text{def}}{=} X_{1i} - X_{2i} \sim N(\mu_1 - \mu_2, \sigma_D^2), \quad i = 1, \ldots, n$

Null hypothesis $H_0: \ \mu_1 - \mu_2 = \Delta_0$ against $H_1: \ \mu_1 - \mu_2 \neq \Delta_0$

Test statistic $T_0 \stackrel{\text{def}}{=} \frac{\bar{D} - \Delta_0}{\frac{S_D}{\sqrt{n}}} \sim t(n-1)$

Test reject H_0 for $|t_0| > t(n-1)_{\alpha/2}$

p-value: $p = 2\left(1 - P(t(n-1) \leq |t_0|)\right)$

CI for $\mu_1 - \mu_2$: $\left[\bar{D} \pm t(n-1)_{\alpha/2} \frac{S_D}{\sqrt{n}}\right]$

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Hypothesis test for comparing variances from population 1 and 2

Model

$$X_{11}, \dots, X_{1n_1} \quad \text{iid} \quad X_{1i} \sim N(\mu_1, \sigma_1^2)$$
$$X_{21}, \dots, X_{2n_2} \quad \text{iid} \quad X_{2i} \sim N(\mu_2, \sigma_2^2)$$

Null hypothesis

$$H_0: \ \sigma_1^2 = \sigma_2^2 \quad \text{against} \quad H_1: \ \sigma_1^2 \neq \sigma_2^2$$

Test statistic

$$F_0 \overset{\text{def}}{=} \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \triangleq \frac{s_1^2}{s_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Test

reject H_0 for $f_0 > F(n_1 - 1, n_2 - 1)_{\alpha/2} \vee f_0 < F(n_1 - 1, n_2 - 1)_{1-\alpha/2}$

p-value: $p = 2\,min\{P(F(n_1 - 1, n_2 - 1) > f_0), P(F(n_1 - 1, n_2 - 1) < f_0)\}$

CI for $\frac{\sigma_1^2}{\sigma_2^2}$:

$$\left[\frac{s_1^2/s_2^2}{F(n_1-1,n_2-1)_{\alpha/2}}, \frac{s_1^2/s_2^2}{F(n_1-1,n_2-1)_{1-\alpha/2}}\right]$$